
Digital Thorium

End-to-End Money Flow Architecture

*From the Saver's Bank Account
to the Reactor Wall
and Back Again*

A companion to the main policy report and diligence memorandum. This document traces every rupee through the Digital Thorium architecture: the primary issuance flow when a saver buys a token, the steady-state quarterly flow when the reactor generates electricity, the secondary trading flow when tokens change hands, and the three independent audit chains that make all of it verifiable.

April 2026

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Reading Guide

The architecture has four distinct money flows, each shown as a separate diagram.

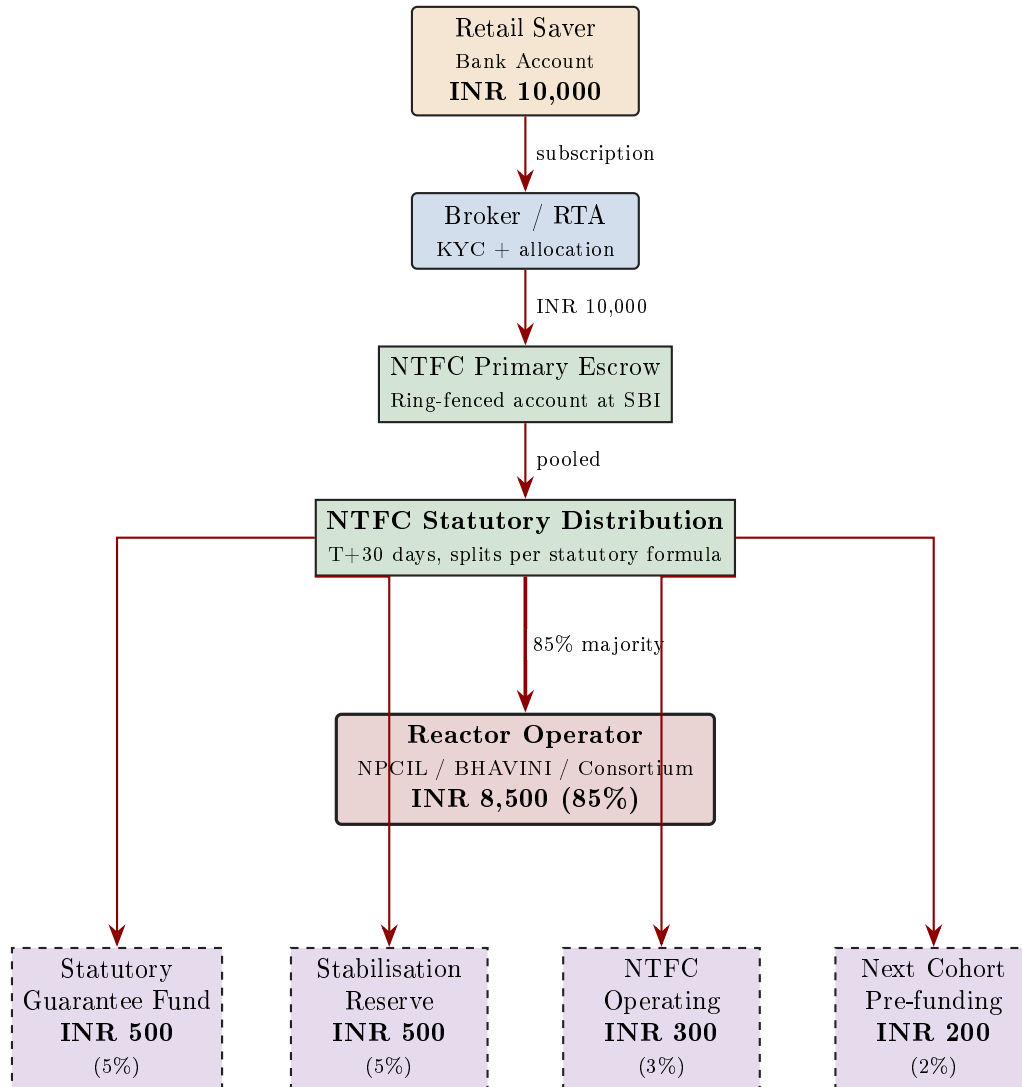
1. **Primary Issuance Flow.** What happens to the saver's INR 10,000 the first time it enters the system. This is the only event at which new cash enters the NTFC perimeter.
2. **Steady-State Quarterly Flow.** What happens every 90 days when the reactor generates electricity, sells it to the grid, and the cash is distributed pro-rata to all token holders.
3. **Secondary Trading Flow.** What happens when one holder sells a token to another. No new cash enters the issuer perimeter; the flow is between holders.
4. **Capital Recycling Flywheel.** How the proceeds from cohort N pay for the construction of cohort N+1 over a multi-year cadence.

A fifth section traces the three independent audit chains that verify each flow. A sixth section walks through a single holder's complete 20-year lifecycle.

1 Primary Issuance Flow

When a saver subscribes to Digital Thorium at primary issuance, the cash takes the following path within 30 days of cohort close. The reference example throughout is an INR 10,000 retail ticket.

Diagram 1. Primary Issuance Flow



What each destination actually does

Destination	INR	Function
Reactor Operator	8,500	NTFC purchases long-dated capacity rights on a commissioned, generating reactor cohort. Operator uses cash to repay project debt, return equity to original sponsors, or re-cycle into next cohort under construction.
Statutory Guarantee Fund	500	Seed for the legally mandated buffer that backstops settlement shortfalls. Held in G-secs and AAA paper. Replenished annually from management fees.
Stabilisation Reserve	500	Working capital that smooths quarterly distributions across capacity-factor variation. Held in short-tenor liquid assets.
NTFC Operating Account	300	Issuance costs: legal, audit, custody, primary dealer underwriting, settlement. Statutorily capped.
Next Cohort Pre-funding	200	Forward purchase of capacity rights on the next reactor cohort under construction. Creates the capital recycling fly-wheel.
Total	10,000	

The architectural point

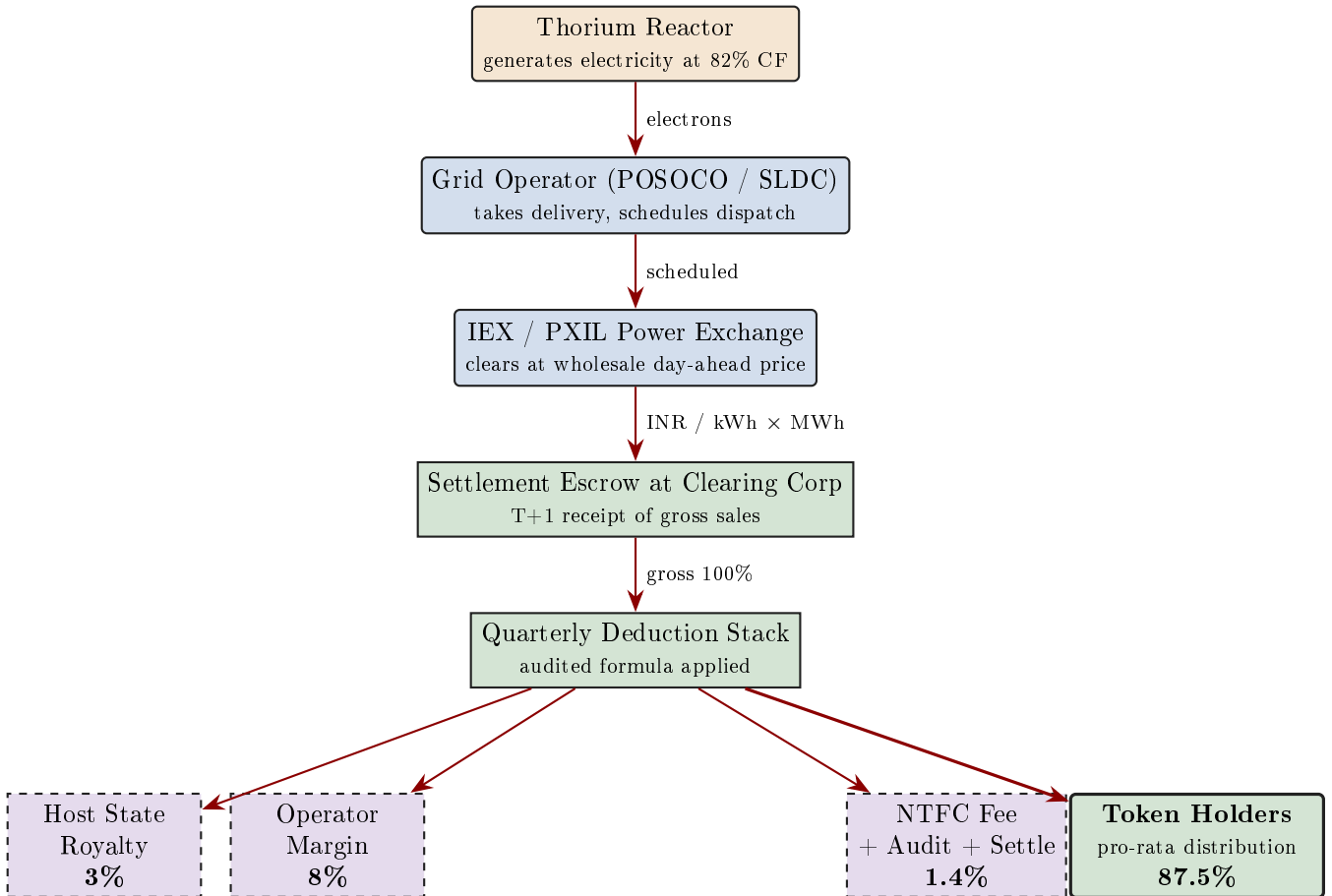
The 85 percent that flows to the operator does not pay for tomorrow's reactor. It pays for yesterday's reactor that has already been built, has commissioned, and is generating electricity on the day of issuance. This is the structural protection against the failure mode of synthetic issuance.

The capital formation effect happens indirectly. By giving the original project lenders and equity sponsors an exit at a market-clearing price, NTFC frees their balance sheets to back the next reactor cohort. The 2030 cohort gets built with capital from sponsors who got their 2026 capital back when the 2026 cohort tokenised.

2 Steady-State Quarterly Flow

Once tokens are outstanding, the reactor generates electricity 24/7. Every 90 days the cash from selling that electricity is distributed to holders according to a published formula. This is the diagram a holder needs to understand to know where the quarterly INR 192 in their bank account comes from.

Diagram 2. Quarterly Cash Distribution



Worked example: one quarter on a 700 MW cohort

For a 700 MW reactor operating at 82 percent capacity factor for 90 days at a wholesale clearing price of INR 5.5 per kWh:

Step	Value	Calculation
Generation (TWh)	1.24	$700 \text{ MW} \times 24 \times 90 \times 0.82$
Gross revenue (INR cr)	682	$1.24 \text{ TWh} \times \text{INR } 5.5/\text{kWh}$
State royalty (INR cr)	20.5	3%
Operator margin (INR cr)	54.6	8%
NTFC fee (INR cr)	6.8	1%
Regulator and settlement (INR cr)	2.7	0.4%
Net to all holders (INR cr)	597	87.5%
Per kWhy outstanding (INR)	4.81	$597 \text{ cr} / 124 \text{ cr kWhy in cohort}$
On a 500 kWhy holding (INR)	2,406	500×4.81
On an INR 10K ticket (INR)	192	at INR 20 launch price per kWhy

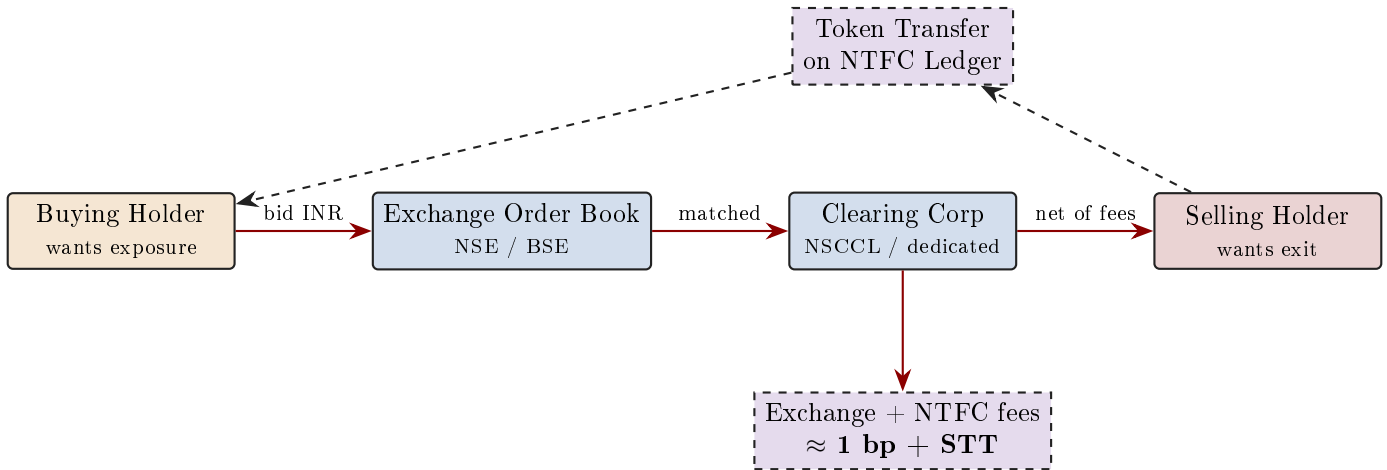
The holder of an INR 10,000 ticket sees approximately INR 192 land in their bank account at the end of the quarter, before any tax friction. Below the proposed TDS threshold so no withholding applies.

Four such payments per year compound the IRR described in the main report.

3 Secondary Trading Flow

When tokens are sold on the exchange, no new cash enters the NTFC perimeter. The flow is between investors, mediated by the clearing corporation. This is identical in structure to selling NTPC equity on NSE.

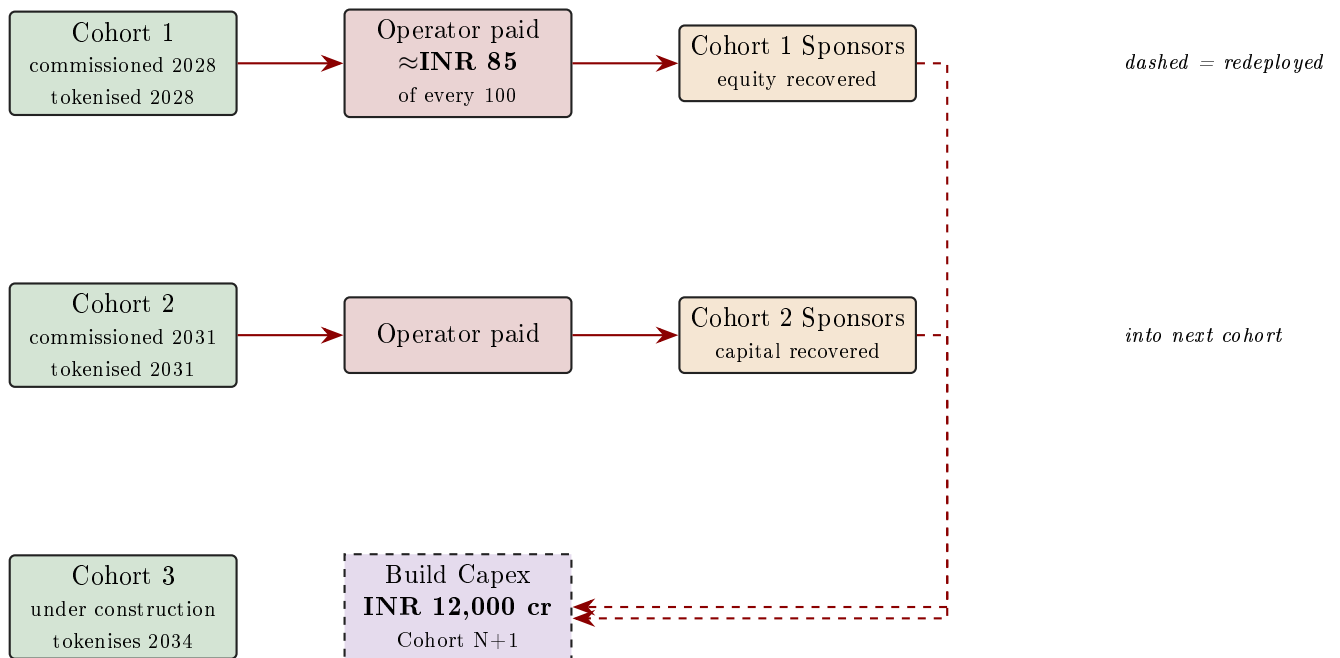
Diagram 3. Secondary Trade



The buyer's cash flows directly to the seller through the clearing corporation. NTFC sees none of it. NTFC and the exchange collect a settlement fee of approximately one basis point on the transaction, plus the standard securities transaction tax goes to the government. The operator sees nothing from secondary trades, which is precisely the property that makes the asset class function like equity rather than like a perpetual fundraising channel.

4 Capital Recycling Flywheel

The most important architectural feature of the system is what happens to the operator's primary proceeds. Tracing this across cohorts shows why Digital Thorium scales without the Union Budget having to write a cheque.

Diagram 4. Multi-Cohort Recycling

The 2028 cohort tokenisation returns capital to sponsors who built it. Those sponsors redeploy into the 2031 cohort. The 2031 tokenisation returns their capital again, and the cycle continues. After three cohorts the system is self-sustaining: each tokenisation event funds the construction of the next.

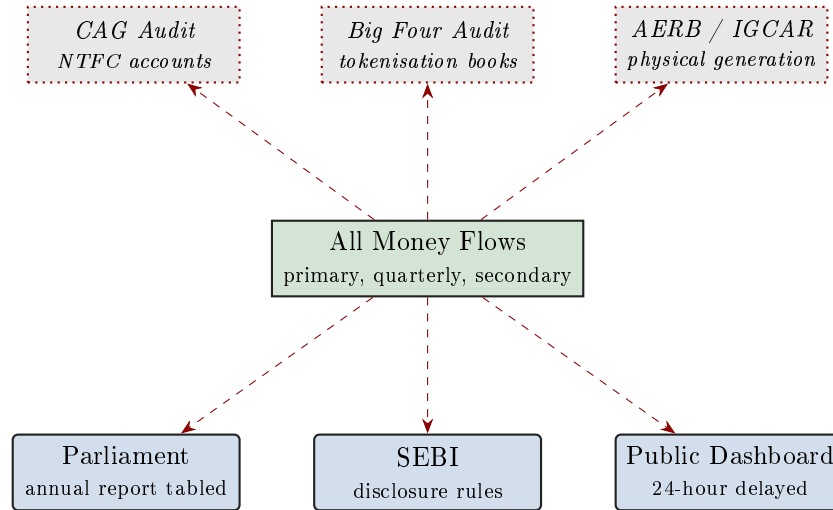
Why this is the right architecture

- **No taxpayer cheque.** The Union Budget is the back-stop, not the funding source. Construction capital comes from sponsors and lenders. Tokenisation returns it. Tokens are bought with household savings.
- **Construction risk stays with sponsors.** The original project lenders and equity holders take build risk. They get paid only on commissioning. Holders of tokens never see construction risk because they never buy uncommissioned capacity.
- **The flywheel accelerates.** Each cohort tokenisation enlarges the pool of capital available for the next cohort. The build rate is not capped by the Union Budget. It is capped only by physical construction capacity and regulatory approval pace.

5 Audit Trail

Every rupee in every diagram above is traceable through three independent audit chains. This is the structural feature that distinguishes Digital Thorium from any failed tokenisation experiment.

Diagram 5. Three Independent Audit Chains



What each chain verifies

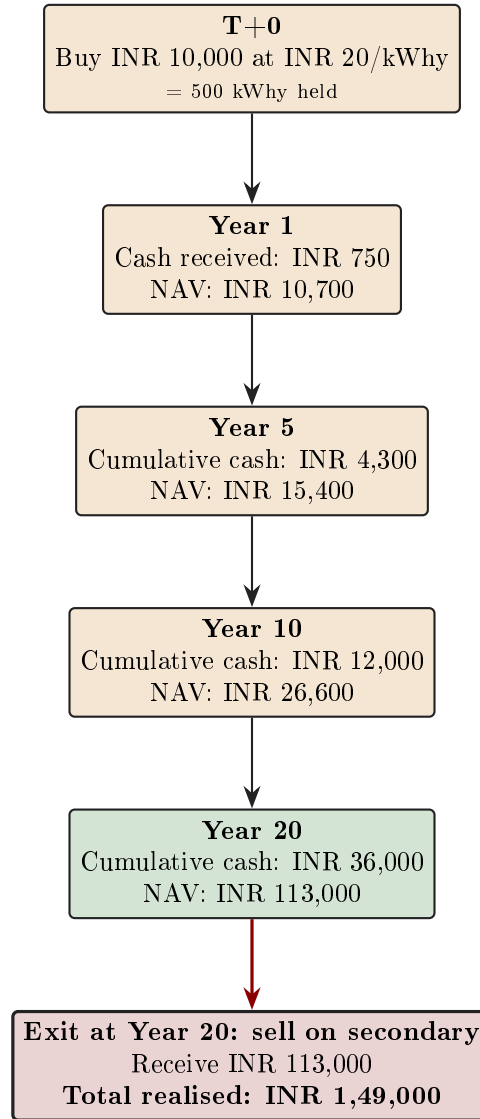
Chain	Verifies	Failure mode it prevents
CAG	NTFC statutory accounts; flow of funds at the corporation level	Misappropriation, off-balance-sheet liabilities, fee drift
Big Four	Tokenisation accounts; per-cohort capacity claims; settlement formula application	Synthetic issuance, dilution events, calculation error
AERB / IGCAR	Physical generation; capacity factor; reactor operating data	Phantom capacity, undisclosed outages, inflated claims
SEBI	Disclosure timeliness; market conduct; primary dealer behaviour	Information asymmetry, market manipulation
Parliament	Annual report tabled; political accountability	Structural drift over electoral cycles
Public dashboard	Real-time visibility for retail holders	Trust erosion through opacity

If any chain detects a discrepancy, the others should also flag it. Three independent chains catching the same problem from different angles is the protection. Crypto tokens fail because no chain audits anything physical. Government programmes fail because no chain audits anything financial. Digital Thorium is engineered to make both kinds of audit unavoidable.

6 Putting It Together: A Single Holder's Lifetime View

To close, a worked example tracing one INR 10,000 ticket from purchase through 20 years of holding to eventual exit.

Diagram 6. Lifetime Flow for One Holder



The holder pays INR 10,000 once at T+0. Over 20 years they receive INR 36,000 in quarterly cash distributions plus a token whose NAV has grown to INR 113,000 on the secondary market. Total realised value: roughly INR 1.49 lakh, before capital gains tax on the appreciation component.

The reactor on the other end of this transaction generated electrons every quarter for 20 years. Those electrons were sold into the grid. The grid sold them to data centres training AI models, to homes running air conditioners, to factories making steel. The cash collected at the wholesale clearing price flowed through the deduction stack and arrived in the holder's bank account every 90 days. The auditors signed off every quarter. The dashboard showed every megawatt-hour.

That is the architecture. Cash enters once. Electrons flow continuously. Distributions flow continuously. Audit verifies continuously. The token represents the holder's slice of this circulating system, and the secondary market prices it in real time.

Notes on this document. Diagrams are schematic. Indicative numbers reflect the central case in the main report. The 85/5/5/3/2 split of primary proceeds is a proposed default and would be set in the NTFC rulebook with the option of cohort-specific variation. The 87.5 percent net pass-through is a steady-state figure; early years may carry higher transaction friction. All figures should be treated as illustrative until the cohort-specific prospectus is published at issuance.